

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399839
Kind Code	B1
Date of Patent	August 26, 2025
Inventor(s)	Patel; Chintan S. et al.

Dynamically controlled cache rinsing

Abstract

The disclosed computer-implemented method includes detecting a low utilization time period for memory traffic between a cache and memory, selecting from the cache, in response to detecting the low utilization time period, a dirty cache entry based on one or more factors, and writing the dirty cache entry back to the memory during the low utilization time period. Various other methods, systems, and computer-readable media are also disclosed.

Inventors: Patel; Chintan S. (Austin, TX), Branover; Alexander J. (Boxborough, MA), Sargunaraj; Joe J. (Santa Clara, CA), Balakrishnan; Ganesh (Austin, TX), Brennan; Christopher J. (Boxborough, MA), Lahiry; Akshay (Boxborough, MA), Kalyanasundharam; Vydhyathan (Santa Clara, CA)

Applicant: Advanced Micro Devices, Inc. (Santa Clara, CA)

Family ID: 1000006899436

Assignee: Advanced Micro Devices, Inc. (Santa Clara, CA)

Appl. No.: 18/090257

Filed: December 28, 2022

Publication Classification

Int. Cl.: G06F12/12 (20160101); G06F12/0891 (20160101); G06F12/123 (20160101); G06F12/128 (20160101)

U.S. Cl.:

CPC G06F12/128 (20130101); G06F12/0891 (20130101); G06F12/123 (20130101);

Field of Classification Search

CPC: G06F (12/12); G06F (12/123); G06F (12/128); G06F (12/08); G06F (12/0891); G06F (12/00)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6625694	12/2002	Masri	711/E12.07	G06F 12/121
8868838	12/2013	Glasco	711/133	G06F 12/126
10681394	12/2019	Field	N/A	H04N 21/231
2008/0244185	12/2007	O'Krafka	711/E12.075	G06F 12/128
2010/0274971	12/2009	Solihin	711/146	G06F 12/0831
2015/0143059	12/2014	Goodman	711/143	G06F 12/0864
2016/0203085	12/2015	Kranich	711/120	G06F 12/0238
2018/0285268	12/2017	Korgaonkar	N/A	G06F 13/4027
2018/0322052	12/2017	Benedict	N/A	G06F 11/3423
2023/0014565	12/2022	Ray	N/A	G06F 12/0811
2024/0143502	12/2023	Dechene	N/A	G06F 9/3836

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2017142562	12/2016	WO	G06F 11/30

OTHER PUBLICATIONS

J. Sim, G. H. Loh, H. Kim, M. OConnor and M. Thottethodi, "A Mostly-Clean DRAM Cache for Effective Hit Speculation and Self-Balancing Dispatch," 2012 45th Annual IEEE/ACM International Symposium on Microarchitecture, Vancouver, BC, Canada, 2012, pp. 247-257. cited by examiner

S. Agarwal and H. K. Kapoor, "Reuse-Distance-Aware Write-Intensity Prediction of Dataless Entries for Energy-Efficient Hybrid Caches," in IEEE Transactions on Very Large Scale Integration (VLSI) Systems, vol. 26, No. 10, pp. 1881-1894, Oct. 2018. cited by examiner

Y. Lee and S. Kim, "RAMS: DRAM Rank-Aware Memory Scheduling for Energy Saving," in IEEE Transactions on Computers, vol. 65, No. 10, pp. 3210-3216, Oct. 1, 2016. cited by examiner

Primary Examiner: Bataille; Pierre Michel

Background/Summary

BACKGROUND

(1) Reducing memory accesses by a processor often improves performance by reducing overhead associated with memory accesses. Processors often include local memory or caches for storing data that is frequently accessed to avoid accessing the memory. Because the processors operate on the cached data, the memory needs to be updated with any changed (e.g., dirty) data in the cache. Cache rinsing involves writing dirty cache entries or lines back to memory before the cache entries are evicted for new entries. Because cache rinsing involves writing to memory, cache rinsing can reduce available memory bandwidth particularly if performed at inopportune times. For example, if the cache rinsing is triggered as part of an eviction for a cache miss, the cache rinsing adds significant overhead to the cache miss.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings illustrate a number of exemplary implementations and are a part of the specification. Together with the following description, these drawings demonstrate and explain various principles of the present disclosure.

(2) FIG. 1 is a block diagram of an exemplary system for dynamically controlled cache rinsing.

(3) FIG. 2 is a block diagram of an example cache hierarchy.

(4) FIG. 3 is a diagram of an exemplary cache rinsing.

(5) FIG. 4 is a flow diagram of an exemplary method for dynamically controlled cache rinsing.

(6) Throughout the drawings, identical reference characters and descriptions indicate similar, but not necessarily identical, elements. While the exemplary implementations described herein are susceptible to various modifications and alternative forms, specific implementations have been shown by way of example in the drawings and will be described in detail herein. However, the exemplary implementations described herein are not intended to be limited to the particular forms disclosed. Rather, the present disclosure covers all modifications, equivalents, and alternatives falling within the scope of the appended claims.

DETAILED DESCRIPTION

(7) The present disclosure is generally directed to dynamically controlled cache rinsing. As will be explained in greater detail below, implementations of the present disclosure opportunistically perform cache rinsing by writing dirty cache entries from a cache back to memory during a low utilization time period between the cache and the memory. The systems and methods provided herein improve the functioning of a computer itself by more efficiently utilizing memory bandwidth and more effectively updating memory coherency, further improving cache performance.

(8) As will be described in greater detail below, the instant disclosure describes various systems and methods for dynamically controlling cache rinsing by detecting an opportunistic time for cache rinsing, selecting dirty cache entries with reduced overhead, and writing back the dirty cache entries.

(9) In one example, a device for dynamically controlled cache rinsing includes a cache and a controller configured to: (1) detect a low utilization time period for memory traffic between a cache and memory, (2) select from the cache, in response to detecting the low utilization time period, a

dirty cache entry based on one or more factors, and (3) write the dirty cache entry back to the memory during the low utilization time period.

(10) In some examples, the one or more factors includes an age of the dirty cache entry. In some examples, selecting the dirty cache entry based on the age prioritizes an older entry.

(11) In some examples, the one or more factors includes a dirtiness of a cache set for the dirty cache entry. In some examples, the dirtiness of the cache set corresponds to a number of dirty cache entries exceeding a threshold number of dirty cache entries. In some examples, selecting the dirty cache entry based on the dirtiness prioritizes a cache set with a larger percentage of dirty cache entries. In some examples, the dirtiness of the cache set corresponds to a number of dirty cache entries exceeding a threshold number of dirty cache entries near eviction and selecting the dirty cache entry based on the dirtiness prioritizes a cache set with a larger number of dirty cache entries near eviction.

(12) In some examples, the one or more factors includes random selection. In some examples, detecting the low utilization time period is asynchronous from a memory request for the cache. In some examples, the method further includes periodically observing the memory traffic between the cache and the memory, wherein detecting the low utilization time period occurs during the periodic observation. In some examples, the low utilization time period corresponds to a number of active operations between the cache to the memory being less than a threshold number of operations.

(13) In one implementation, a system for dynamically controlled cache rinsing includes a physical memory, a cache corresponding to the physical memory, and a controller configured to observe, asynchronously from a memory request for the cache, a memory traffic between the cache and the memory, detect, during the observation, a low utilization time period for the memory traffic, select from the cache, in response to detecting the low utilization time period, a dirty cache entry based on one or more factors, and write the dirty cache entry back to the memory during the low utilization time period.

(14) In some examples, the one or more factors includes an age of the dirty cache entry and selecting the dirty cache entry prioritizes an older entry. In some examples, the one or more factors includes a dirtiness of a cache set for the dirty cache entry that corresponds to a number of dirty cache entries exceeding a threshold number of dirty cache entries and selecting the dirty cache entry prioritizes a cache set with a larger percentage of dirty cache entries. In some examples, the one or more factors includes random selection.

(15) In some examples, the low utilization time period corresponds to a number of active operations between the cache to the memory being less than a threshold number of operations.

(16) In one implementation, a method for dynamically controlled cache rinsing includes (1) detecting a low utilization time period for a write traffic from the cache, (2) selecting from the cache, in response to detecting the low utilization time period, a dirty cache entry based on an age of the dirty cache entry, and (3) writing back the dirty cache entry during the low utilization time period.

(17) In some examples, selecting the dirty cache entry prioritizes an older entry. In some examples, selecting the dirty cache entry is further based on a dirtiness of a cache set for the dirty cache entry, and the dirtiness of the dirty cache entry corresponds to a number of dirty cache entries exceeding a threshold number of dirty cache entries. In some examples, selecting the dirty cache entry prioritizes a cache set with a larger percentage of dirty cache entries.

(18) In some examples, the method further includes periodically observing the write traffic from the cache. In some examples, detecting the low utilization time period occurs during the periodic observation. In some examples, detecting the low utilization time period is asynchronous from a memory request for the cache. In some examples, the low utilization time period corresponds to a number of active operations for the cache being less than a threshold number of operations.

(19) Features from any of the implementations described herein can be used in combination with one another in accordance with the general principles described herein. These and other

implementations, features, and advantages will be more fully understood upon reading the following detailed description in conjunction with the accompanying drawings and claims.

(20) The following will provide, with reference to FIGS. **1-4**, detailed descriptions of dynamically controlled cache rinsing. Detailed descriptions of example systems for dynamically controlled cache rinsing will be provided in connection with FIG. **1**. Detailed descriptions of an example cache hierarchy will be provided in connection with FIG. **2**. Detailed descriptions of exemplifying cache rinsing will be provided in connection with FIG. **3**. Detailed descriptions of corresponding computer-implemented methods will also be provided in connection with FIG. **4**.

(21) FIG. **1** is a block diagram of an example system **100** for dynamically controlled cache rinsing. System **100** corresponds to a computing device, such as a desktop computer, a laptop computer, a server, a tablet device, a mobile device, a smartphone, a wearable device, an augmented reality device, a virtual reality device, a network device, and/or an electronic device. As illustrated in FIG. **1**, system **100** includes one or more memory devices, such as memory **120**. Memory **120** generally represents any type or form of volatile or non-volatile storage device or medium capable of storing data and/or computer-readable instructions. Examples of memory **120** include, without limitation, Random Access Memory (RAM), Read Only Memory (ROM), flash memory, Hard Disk Drives (HDDs), Solid-State Drives (SSDs), optical disk drives, caches, variations or combinations of one or more of the same, and/or any other suitable storage memory.

(22) As illustrated in FIG. **1**, example system **100** includes one or more physical processors, such as processor **110**. Processor **110** generally represents any type or form of hardware-implemented processing unit capable of interpreting and/or executing computer-readable instructions. In some examples, processor **110** accesses and/or modifies data and/or instructions stored in memory **120**. Examples of processor **110** include, without limitation, microprocessors, microcontrollers, Central Processing Units (CPUs), graphics processing units (GPUs), Field-Programmable Gate Arrays (FPGAs) that implement softcore processors, Application-Specific Integrated Circuits (ASICs), systems on chip (SoCs), digital signal processors (DSPs), Neural Network Engines (NNEs), accelerators, graphics processing units (GPUs), portions of one or more of the same, variations or combinations of one or more of the same, and/or any other suitable physical processor.

(23) As further illustrated in FIG. **1**, processor **110** includes a core **112**, a cache **114**, and a controller **142**. Core **112** corresponds to a processor core, although in other examples corresponds to a chiplet such as a chiplet of an accelerator, or any other processing unit, Cache **114** corresponds to a cache used by processor **110** (e.g., a client-side cache such as a low-level cache or L1 cache). In some examples, cache **114** corresponds to and/or includes other caches, such as a memory-side cache. Controller **142** corresponds to a control circuit that can manage cache **114**, such as by performing cache rinsing. In some examples, controller **142** also controls aspects of cache **114**.

(24) Processor **110** reads and operates on instructions and/or data stored in memory **120**. Because memory **120** is often slower than processor **110**, memory access times create bottlenecks for processor **110**. To alleviate this problem, processor **110** includes cache **114**, which is typically a fast memory with access times less than that of memory **120**, in part due to being physically located in processor **110**.

(25) Cache **114** holds data and/or instructions read from memory **120**. Processor **110** (and/or core **112**) first makes memory requests to cache **114**. If cache **114** holds the requested data (e.g., a cache hit), processor **110** reads the data from cache **114** and avoids the memory access times of memory **120**. If cache **114** does not hold the requested data (e.g., a cache miss), processor **110** retrieves the data from memory **120**, incurring the memory access time. Although a larger cache size can reduce cache misses, considerations such as die size and power consumption limits the size of cache **114**. Thus, to further reduce the need to access memory **120** on cache misses, processor **110** can incorporate another cache (e.g., one or more intervening levels of caches), that is larger but slower than cache **114**, in a cache hierarchy.

(26) As will be described further below, as processor **110** operates on data stored in cache **114**, the

modified or dirty data entries must be written back to memory **120** to maintain the coherency of memory **120**. The dirty entries can be written back immediately prior to eviction (e.g., clearing of entries in cache **114** to make space for new entries), at the cost of adding overhead to the eviction. The cache rinsing described herein provides for an opportunistic approach.

(27) FIG. 2 illustrates an example cache hierarchy in a system **200** which corresponds to system **100**. System **200** includes one or more processors **210** which corresponds to processor **110** and one or more accelerators **211** which corresponds to processor **110**. As illustrated in FIG. 2, processor **210** includes a core **212A** which corresponds to core **112**, a core **212B** which corresponds to core **112**, an L1 cache **214A** which corresponds to cache **114**, an L1 cache **214B** which corresponds to cache **114**, an L2 cache **216A** which can correspond to cache **114**, an L2 cache **216B** which can correspond to cache **114**, and an L3 cache **218** which can correspond to cache **114**.

(28) In the cache hierarchy of FIG. 2, level 1 (L1) corresponds to a lowest level of the hierarchy. L1 caches, such as L1 cache **214A** and L1 cache **214B**, can be implemented with a fast memory, such as static random-access memory (SRAM). To further prioritize speed, L1 caches can also be integrated with processor **210**, for example within core **212A** and core **212B** respectively, which can improve latency and throughput. In some examples, as shown in FIG. 2, processor **210** includes multiple L1 caches.

(29) L2 caches, such as L2 cache **216A** and L2 cache **216B**, are the next level in the cache hierarchy after L1 caches, which can be larger than and slower than L1 caches. Although integrated with processor **210**, L2 caches can, in some examples, be located outside of a chip core, but can also be located on the same chip core package. L3 caches such as L3 cache **218** can be larger than L2 caches but can also be slower. In some examples, L3 caches can serve as a bridge to the main memory (e.g., memory **220**). As such, L3 caches can be faster than the main memory. In some examples, multiple processors and/or cores can share an L3 cache, which can be located on the same chip core package or outside the package.

(30) Memory **220** which corresponds to memory **120**, stores instructions and/or data for processor **210** to read and use. Memory **220** can be implemented with dynamic random-access memory (DRAM). As shown in FIG. 2, the cache hierarchy further includes a memory cache **230** (e.g., a memory-side cache) which in some examples corresponds to cache **114**, and a data fabric **240** which corresponds to various structures, connections, and control circuits for sending data between memory and cache structures.

(31) System **200** also includes one or more accelerators having a similar cache hierarchy. Accelerator **211** includes a chiplet **213A** which corresponds to core **112**, a chiplet **213B** which corresponds to core **112**, a chiplet **213C** which corresponds to core **112**, a chiplet **213D** which corresponds to core **112**, and an L2 cache **217** which corresponds to cache **114** that is shared by the chiplets.

(32) Available bandwidth between the caches and the memory reduces as the memory traffic between the caches and the memory increases. For example, as cache misses occur (e.g., requested data not in cache), the caches must fetch the requested data from the memory, reducing the bandwidth. If an eviction is needed to fetch the requested data, and there is dirty data to be written back to the memory, the write back also reduces the bandwidth. As will be described further below, the caches can update the respective memories via cache rinsing to reduce the write backs.

(33) FIG. 3 illustrates a system **300** corresponding to system **100**. System **300** includes a cache **314** corresponding to cache **114** and a memory **320** corresponding to memory **120**. As further illustrated in FIG. 3, cache **314** includes various lines or entries for holding data, such as a cache entry **350A**, a cache entry **350B**, a cache entry **350C**, a cache entry **350D**, a cache entry **350E**, a dirty cache entry **352A**, a dirty cache entry **352B**, and a dirty cache entry **352C**. In some examples, the cache entries correspond to lines. The cache entries can be organized into sets, such as cache entry **350A**, dirty cache entry **352A**, cache entry **350B**, and cache entry **350C** being in cache set **370A**, and dirty cache entry **352B**, cache entry **350D**, dirty cache entry **352C**, and cache entry **350E** being in cache

set **370B**. In some examples, the cache sets correspond to pages. Memory **320** includes various lines or entries as well (e.g., addresses), such as a memory entry **360A**, a memory entry **360B**, a memory entry **360C**, a memory entry **360D**, a memory entry **360E**, a memory entry **360F**, a memory entry **360G**, a memory entry **360H**, etc.

(34) Dirty cache entries, such as dirty cache entry **352A**, dirty cache entry **352B**, and/or dirty cache entry **352C**, correspond to data that has been modified (e.g., written to) since being read from memory **320**. In some examples, cache **314** and/or a controller thereof (e.g., controller **142**) tracks whether entries are dirty, for instance by storing metadata of cache entry properties. During an eviction **354**, the controller can select dirty cache entry **352A** according to an appropriate eviction policy (e.g., selecting a page or cache set **370A** including dirty cache entry **352A** for eviction). The controller can load the metadata for dirty cache entry **352A** to determine whether dirty cache entry **352A** should be written back to memory **320** (e.g. to its corresponding address at memory entry **360C**) before deleting from cache **314**.

(35) In some examples, performing the write back to memory during eviction **354** can add overhead to a memory request that triggered eviction **354**. For instance, the controller loads the metadata in an operation having overhead, and writing data back to memory **320** consumes bandwidth. To mitigate such overhead, cache rinsing can occur before eviction.

(36) Rinse **356** occurs prior to dirty cache entry **352B** being selected for eviction. For instance, rather than waiting for an eviction involving dirty cache entry **352B** to write back the dirty data to memory entry **360D**, the controller can proactively (e.g., asynchronously from an eviction and/or memory request) write back or rinse the dirty data. In some implementations, the controller can wait for or detect when bandwidth is available. For example, the controller can periodically check whether the available bandwidth is above a bandwidth threshold, which can correspond to a low utilization time period of memory traffic. In some examples, the low utilization time period corresponds to a number of active and/or pending memory requests between cache **314** and memory **320** (e.g., read and/or write requests) are below a threshold number of memory requests. In some examples, the low utilization time period corresponds to the number of active and/or pending memory requests being within a threshold range for a threshold period of time (e.g., number of cycles). In some implementations, the controller can actively detect when there is a low utilization time period. Thus, when the controller detects sufficient available bandwidth, the controller can perform rinse **356**. Further, in some implementations, rinse **356** includes rinsing all cache entries of the respective cache set (e.g., rinsing dirty cache entry **352B** includes also rinsing cache entry **350D**, dirty cache entry **350C**, and cache entry **350E**).

(37) In some implementations, the controller selects between candidate dirty cache entries based on one or more factors. For example, the controller can select dirty cache entry **352C** before dirty cache entry **352B** to perform a rinse **358** (of writing the data from dirty cache entry **352C** back to memory entry **360G**) before rinse **356**. Some of the factors include dirtiness (e.g., an amount of dirty data) and age (e.g., how long data has been in cache **314**).

(38) In some examples, dirtiness can be considered as whether a cache set is dirty (e.g., has dirty data), has a sufficient amount of dirty data (e.g., having a number of dirty cache lines exceeding a threshold number of dirty cache lines), or compared between cache sets. In some examples, dirtiness can correspond to a number of non-dirty cache entries in the respective cache set, which can correspond to a number of non-dirty cache entries that will be evicted. In some examples, the dirtiness can correspond to a number of non-evicting cache entries (e.g., cache entries that are not dirty or cache entries that are not within N, a threshold number of misses, away from eviction) that will be evicted.

(39) In some examples, age can be considered based on how long the data has been in cache **314**, or how long since the data was last used (e.g., tracked as least recently used (LRU)). In some examples, age can correspond to an eviction age, such as a maximum or minimum time to live. In other examples, age can correspond to a number of misses.

(40) In some implementations, the controller can select the best dirty cache entry candidate for rinsing first. For instance, a dirtier cache set (e.g., having a larger number and/or percentage of dirty cache lines) can be prioritized first, and older entries can be prioritized second. For example, in FIG. 3, cache set **370B** can be selected over cache set **370A** based on dirtiness (e.g., cache set **370B** has more dirty cache entries and/or a higher percentage of dirty cache entries than cache set **370A**, or cache set **370B** has fewer non-dirty cache entries than cache set **370A**). From the selected cache set, the oldest dirty cache entry can be selected. Thus, the dirtiest, oldest cache entry can be selected. However, in some examples, reading the relevant metadata and comparing candidates can introduce undesirable overhead. For example, determining the dirtiness can require fetching metadata. As such, in other implementations, the controller can rely on more easily accessible (e.g., quickly fetchable) factors.

(41) In some implementations, the controller can more readily determine age than dirtiness. For instance, the controller can more quickly read LRU metadata than dirtiness metadata. Thus, the controller can select a dirty cache entry based on age. In some examples, the controller can rely on other factors, such as to select between similarly aged cache entries. In some examples, the controller can randomly select between candidates. For instance, rather than fetching dirtiness metadata, the controller can more quickly randomly select between similarly aged cache entries.

(42) In yet other implementations, the controller can rely on any combination and prioritization thereof of factors determined from the cache entry metadata. In some examples, the controller can select which factors to prioritize based on available bandwidth. For instance, when more bandwidth is available, the controller can prioritize selecting the best candidates even if the selection process incurs overhead. When less bandwidth is available, the controller can prioritize faster candidate selection.

(43) Moreover, in some implementations, to avoid unneeded writing of data to memory **320**, cache rinsing can be suppressed even if memory bandwidth is available. If the factors described herein do not produce a desirable candidate for rinsing, the controller can wait for the next low utilization time period. For example, if a number of dirty cache entries have not hit the threshold (e.g., within a cache set) a cache set may not be selected. In addition, after selecting a cache set if an age of a dirty cache entry is not close to an eviction age (e.g., does not satisfy an eviction age threshold), the controller can skip the dirty cache entry for rinsing. In another example, the controller can prioritize cache sets based on a number of non-dirty and/or non-evicting cache entries lines exceeding N misses away from eviction until no cache sets are above the threshold number of cache entries.

(44) FIG. 4 is a flow diagram of an exemplary computer-implemented method **400** for dynamically controlled cache rinsing. The steps shown in FIG. 4 can be performed by any suitable computer-executable code and/or computing system, including the system(s) illustrated in FIGS. 1 and/or 2. In one example, each of the steps shown in FIG. 4 represent an algorithm whose structure includes and/or is represented by multiple sub-steps, examples of which will be provided in greater detail below.

(45) As illustrated in FIG. 4, at step **402** one or more of the systems described herein detect a low utilization time period for memory traffic between a cache and memory. For example, controller **142** can detect a low utilization time period for memory traffic between cache **114** and memory **120**.

(46) The systems described herein can perform step **402** in a variety of ways. In one example, detecting the low utilization time period is asynchronous from a memory request for the cache. Controller **142** can periodically wake up and observe the memory traffic between cache **114** and memory **120**. In some examples, the low utilization time period corresponds to a number of active operations between cache **114** to memory **120** being less than a threshold number of operations. Controller **142** can detect the low utilization time period during the periodic observation.

(47) At step **404** one or more of the systems described herein select from the cache, in response to

detecting the low utilization time period, a dirty cache entry based on one or more factors. For example, controller **142** selects a dirty cache entry based on at least one factor.

(48) The systems described herein can perform step **404** in a variety of ways. In one example, controller **142** can read metadata for cache entries for analyzing the factors. The one or more factors can include an age of the dirty cache entry such that selecting the dirty cache entry based on the age prioritizes an older entry.

(49) The one or more factors can include a dirtiness of a cache set for the dirty cache entry such that selecting the dirty cache entry based on the dirtiness prioritizes a cache set with a larger percentage of dirty cache entries. In some examples, the dirtiness of the cache set corresponds to a number of dirty cache entries exceeding a threshold number of dirty cache entries. In some examples, the dirtiness of the cache set corresponds to a number of dirty cache entries exceeding a threshold number of dirty cache entries near eviction such that selecting the dirty cache entry based on the dirtiness prioritizes a cache set with a larger number of dirty cache entries near eviction. A dirty cache entry near eviction can correspond to a dirty cache entry closer than N (e.g., an eviction threshold) number of misses away from eviction. In some examples, the dirtiness of the cache set corresponds to a number of non-dirty cache entries and/or a number of non-evicting cache entries (e.g., non-dirty cache entries and dirty cache entries that are not near eviction).

(50) The one or more factors can also include random selection (e.g., as a tiebreaker), such as random selection between cache entries similarly evaluated based on other factors.

(51) At step **406** one or more of the systems described herein write the dirty cache entry back to the memory during the low utilization time period. For example, controller **142** can write the dirty data from the selected dirty cache entry back to memory **120** during the low utilization time period. Accordingly, controller **142** can opportunistically perform cache rinsing during times of available bandwidth between cache **114** and memory **120**.

(52) As detailed above, the systems and method described herein provide a dynamic technique for rinsing a cache. Cache rinsing is a technique in which cache lines are written back to memory opportunistically prior to an eviction occurring. During system operation, a cache controller searches for periods of time in which the cache is relatively under-used. During these periods of time, the cache controller writes back the data in dirty lines to the memory. The cache controller selects cache lines according to a set of selection criteria to maximize impact. One such criterion is the number of dirty cache lines in a set. For any particular set, if the number of dirty cache lines is above a threshold, the cache controller selects the dirty lines of that set for write back.

(53) For any particular cache, the bandwidth to the cache is determined by certain factors. If requested data is in the cache such that mostly hits occur, then the highest bandwidth is achieved. If requested data is not in the cache, then misses occur and the cache fetches the requested data from memory and the bandwidth is lowered. If misses occur and the cache includes dirty lines, then an even lower amount of bandwidth is achieved, since some of the bandwidth between the memory and the cache must be expended on performing write-backs.

(54) The present disclosure provides for opportunistically checking for cache lines to rinse during operation. While the system is running, a processor accesses the cache. In certain periods of time, such as where the processor accesses data mostly already stored in its internal cache, the cache is accessed very lightly. While the cache is accessed lightly, bandwidth between the cache and memory exists. Thus, during these periods, a cache controller is able to rinse the cache. Rinsing the cache improves bandwidth during high utilization periods if the cache has a large number of dirty lines because evicting dirty lines requires writing the data in those dirty lines back to the memory, which consumes some bandwidth. By writing back dirty cache lines during low utilization periods, bandwidth is freed for use during high utilization periods. The dynamic cache rinsing described herein includes selecting cache lines for rinsing based on certain criteria, such as the percentage of cache lines in a set that are dirty.

(55) The dynamic cache rinsing can improve cache performance by improving cache bandwidth

during high utilization periods of time. More specifically, without dynamic cache rinsing as described herein, cache bandwidth can be greatly reduced during high utilization periods of time if a large number of write-backs to a backing memory occurs. Using the low utilization periods of time to perform such write-backs frees up bandwidth for other uses during the high utilization periods of time.

(56) As detailed above, the computing devices and systems described and/or illustrated herein broadly represent any type or form of computing device or system capable of executing computer-readable instructions, such as those contained within the modules described herein. In their most basic configuration, these computing device(s) each include at least one memory device and at least one physical processor.

(57) In some examples, the term “memory device” generally refers to any type or form of volatile or non-volatile storage device or medium capable of storing data and/or computer-readable instructions. In one example, a memory device stores, loads, and/or maintains one or more of the modules and/or circuits described herein. Examples of memory devices include, without limitation, Random Access Memory (RAM), Read Only Memory (ROM), flash memory, Hard Disk Drives (HDDs), Solid-State Drives (SSDs), optical disk drives, caches, variations or combinations of one or more of the same, or any other suitable storage memory.

(58) In some examples, the term “physical processor” generally refers to any type or form of hardware-implemented processing unit capable of interpreting and/or executing computer-readable instructions. In one example, a physical processor accesses and/or modifies one or more modules stored in the above-described memory device. Examples of physical processors include, without limitation, microprocessors, microcontrollers, Central Processing Units (CPUs), Field-Programmable Gate Arrays (FPGAs) that implement softcore processors, Application-Specific Integrated Circuits (ASICs), systems on a chip (SoCs), digital signal processors (DSPs), Neural Network Engines (NNEs), accelerators, graphics processing units (GPUs), portions of one or more of the same, variations or combinations of one or more of the same, or any other suitable physical processor.

(59) In some implementations, the term “computer-readable medium” generally refers to any form of device, carrier, or medium capable of storing or carrying computer-readable instructions. Examples of computer-readable media include, without limitation, transmission-type media, such as carrier waves, and non-transitory-type media, such as magnetic-storage media (e.g., hard disk drives, tape drives, and floppy disks), optical-storage media (e.g., Compact Disks (CDs), Digital Video Disks (DVDs), and BLU-RAY disks), electronic-storage media (e.g., solid-state drives and flash media), and other distribution systems.

(60) The process parameters and sequence of the steps described and/or illustrated herein are given by way of example only and can be varied as desired. For example, while the steps illustrated and/or described herein are shown or discussed in a particular order, these steps do not necessarily need to be performed in the order illustrated or discussed. The various exemplary methods described and/or illustrated herein can also omit one or more of the steps described or illustrated herein or include additional steps in addition to those disclosed.

(61) The preceding description has been provided to enable others skilled in the art to best utilize various aspects of the exemplary implementations disclosed herein. This exemplary description is not intended to be exhaustive or to be limited to any precise form disclosed. Many modifications and variations are possible without departing from the spirit and scope of the present disclosure. The implementations disclosed herein should be considered in all respects illustrative and not restrictive. Reference should be made to the appended claims and their equivalents in determining the scope of the present disclosure.

(62) Unless otherwise noted, the terms “connected to” and “coupled to” (and their derivatives), as used in the specification and claims, are to be construed as permitting both direct and indirect (i.e., via other elements or components) connection. In addition, the terms “a” or “an,” as used in the

specification and claims, are to be construed as meaning “at least one of.” Finally, for ease of use, the terms “including” and “having” (and their derivatives), as used in the specification and claims, are interchangeable with and have the same meaning as the word “comprising.”

Claims

1. A method comprising: detecting a low utilization time period for a write traffic from a cache; prioritizing one or more factors of a plurality of factors that includes at least one of an age of a dirty cache entry that prioritizes selecting an older entry, a dirtiness of a cache set of the cache, or a random selection based on an available bandwidth and an overhead associated with each of the plurality of factors; selecting from the cache set of the cache, in response to detecting the low utilization time period, the dirty cache entry based on the prioritized one or more factors; and writing back the dirty cache entry during the low utilization time period.
2. The method of claim 1, wherein the dirtiness of the cache set corresponds to a number of dirty cache entries exceeding a threshold number of dirty cache entries.
3. The method of claim 2, wherein selecting the dirty cache entry prioritizes a cache set with a larger percentage of dirty cache entries.
4. The method of claim 1, further comprising periodically observing the write traffic from the cache, wherein: detecting the low utilization time period occurs during the periodic observation; detecting the low utilization time period is asynchronous from a memory request for the cache; and the low utilization time period corresponds to a number of active operations for the cache being less than a threshold number of operations.
5. The method of claim 1, further comprising suppressing selection of the dirty cache entry when the cache set does not reach a threshold number of dirty cache entries or the age of the dirty cache entry does not satisfy an eviction age threshold.
6. The method of claim 1, the age of the dirty cache or the random selection is prioritized when less bandwidth is available, and the dirtiness is prioritized when more bandwidth is available.
7. A device comprising: a cache; and a controller configured to: detect a low utilization time period for memory traffic between a cache and memory; prioritize one or more factors of a plurality of factors that includes at least one of an age of a dirty cache entry that prioritizes selecting an older entry, a dirtiness of a cache set of the cache, or a random selection based on an available bandwidth and an overhead associated with each of the plurality of factors; select from the cache set of the cache, in response to detecting the low utilization time period, the dirty cache entry based on the prioritized one or more factors; and write the dirty cache entry back to the memory during the low utilization time period.
8. The device of claim 7 wherein the dirtiness of the cache set corresponds to a number of dirty cache entries exceeding a threshold number of dirty cache entries and selecting the dirty cache entry based on the dirtiness prioritizes a cache set with a larger percentage of dirty cache entries.
9. The device of claim 7 wherein the dirtiness of the cache set corresponds to a number of dirty cache entries exceeding a threshold number of dirty cache entries near eviction and selecting the dirty cache entry based on the dirtiness prioritizes a cache set with a larger number of dirty cache entries near eviction.
10. The device of claim 7, wherein detecting the low utilization time period is asynchronous from a memory request for the cache.
11. The device of claim 7, wherein the controller is configured to periodically observe the memory traffic between the cache and the memory, wherein detecting the low utilization time period occurs during the periodic observation.
12. The device of claim 7, wherein the low utilization time period corresponds to a number of active operations between the cache to the memory being less than a threshold number of operations.

13. The device of claim 7, wherein the controller is configured to suppress selecting the dirty cache entry when the cache set does not reach a threshold number of dirty cache entries.
 14. The device of claim 7, wherein the controller is configured to suppress selecting the dirty cache entry when the age of the dirty cache entry does not satisfy an eviction age threshold.
 15. The device of claim 7, wherein the controller is configured to prioritize the age of the dirty cache or the random selection when less bandwidth is available, and prioritize the dirtiness of the cache set when more bandwidth is available.
 16. A system comprising: a physical memory; a cache corresponding to the physical memory; and a controller configured to: observe, asynchronously from a memory request for the cache, a memory traffic between the cache and the memory; detect, during the observation, a low utilization time period for the memory traffic; prioritize one or more factors of a plurality of factors that includes at least one of an age of a dirty cache entry that prioritizes selecting an older entry, a dirtiness of a cache set of the cache, or a random selection based on an available bandwidth and an overhead associated with each of the plurality of factors; select from the cache set of the cache, in response to detecting the low utilization time period, the dirty cache entry based on the prioritized one or more factors; and write the dirty cache entry back to the memory during the low utilization time period.
 17. The system of claim 16, wherein the dirtiness of the cache set corresponds to a number of dirty cache entries exceeding a threshold number of dirty cache entries and selecting the dirty cache entry prioritizes a cache set with a larger percentage of dirty cache entries.
 18. The system of claim 16, wherein the low utilization time period corresponds to a number of active operations between the cache to the memory being less than a threshold number of operations.
 19. The system of claim 16, wherein the controller is configured to suppress selecting the dirty cache entry when the cache set does not reach a threshold number of dirty cache entries or the age of the dirty cache entry does not satisfy an eviction age threshold.
 20. The system of claim 16, wherein the controller is configured to prioritize the age of the dirty cache or the random selection when less bandwidth is available, and prioritize the dirtiness of the cache set when more bandwidth is available.
-